



Louey Barbirou

Data Scientist / ML Engineer apprenticeship · Available immediately

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Profile

Third-year AI & Data student at Ynov Aix-en-Provence. I take a data project from raw data all the way to production, thinking about maintenance, performance and scaling. My **WattCast** model forecasts next-day French electricity demand with a lower error than the official forecast from RTE. After a 4-month AI internship at triTon, I am looking for a **data / AI apprenticeship** (2 weeks in the company / 1 week at school).

Work experience

- AI & Data Development Intern** · triTon, Aix-en-Provence April – July 2026
- Development of **VideoInsight**, a web app for AI video analysis and translation: transcription (Whisper), translation (MarianMT), voice-cloning dubbing (XTTS), lip sync and automatic scene splitting.
 - Preparation pipelines for the audio/video datasets (cleaning, structuring) used to train and evaluate the models.
 - Quality dashboard (transcription and translation accuracy, processing time) to monitor product performance.
- Stock Manager & Flow Analyst** · Novogel, Sfax (Tunisia) 2023
- Forecast of supply needs from sales history; tracking of stock rotation KPIs to reduce waste.
- Head chef, commis, pizza chef** · Paupiette Groupe & Junior's (Marseille), Italian Trattoria (Aix) 2021 – 2026
- Menu creation, stock management and rush service up to 400 covers, team coordination under pressure.

Key projects

Code, repositories and live demos: baluva.github.io/portfolio

WattCast · Electricity demand forecasting in production — daily model forecasting next-day French electricity demand, scored against RTE's official forecast: automated pipeline, dbt warehouse, leakage-free walk-forward backtest 2023-2026, drift monitoring and retraining. **1.90% MAPE vs 1.97% for RTE**.

Stack: Python, LightGBM, dbt, DuckDB, GitHub Actions, FastAPI, MLflow, Evidently · Code: github.com/baluva/wattcast

SupplyPulse · Supply chain analytics — 180,000 real orders (DataCo): layered warehouse (raw, staging, marts) on DuckDB, data dictionary and automated quality tests in CI, decision dashboard. **54.8% late deliveries** surfaced, ~10.8% margin.

Stack: Python, DuckDB, SQL, Streamlit, pandas · Code: github.com/baluva/supplypulse

Prompt Booster · AI prompt optimizer — you paste a quickly written prompt and the tool rewrites it as a structured prompt (role, task, context, constraints, output format). Rule-based Python engine, no API key and no data sent anywhere: task detection (9 types) and FR/EN language detection, context questions to fill in, **score out of 100 before / after** (7 weighted criteria), Markdown, XML or plain-text export. Custom Streamlit UI, CLI, 33 pytest tests in CI. Building it is how I learned prompt engineering.

Stack: Python, Streamlit, pytest, GitHub Actions · Code: github.com/baluva/prompt-booster

Bac Quiz · AI-powered exam revision — automatic multiple-choice question generation from PDF exam papers with multimodal (vision) AI, schema validation and a real-time backend. Live app used by Tunisian high-school students.

Stack: Python, multimodal AI, Pydantic, Supabase, React · Live demo: bacquiz-tn.netlify.app · Code: github.com/baluva/bac-quiz

Olist · E-commerce decision dashboard — repeat-purchase and delivery-time KPIs, BI dashboard designed following Few and Tufte's data visualization principles.

Stack: Python, SQL, PostgreSQL, Apache Superset, Docker

Phishing detection · Email classification — NLP models detecting malicious emails, on an industrial problem statement from Thales.

Stack: Python, scikit-learn, NLP

Also in production: Dbug Hunter, bug bounty assistant (louey9999-dbug-hunter.hf.space) · Code Route TN, Tunisian driving-test quiz (code-route-tn.pages.dev) · Magic Autos, showcase website for a car dealership ([see the site](https://see-the-site)). **Other:** Swarm Prediction (crop yield, FAO & World Bank data).

Skills

Languages: Python, SQL, R, SAS, JavaScript

ML & data science: pandas, NumPy, scikit-learn, XGBoost, LightGBM, forecasting, statistics, time-based validation

AI & NLP: LLMs (Claude, Gemini), multimodal AI, Whisper, MarianMT, XTTS, Hugging Face, Ollama, prompt engineering

Data engineering & BI: dbt, DuckDB, PostgreSQL, MongoDB, Supabase, ETL, Apache Superset, Streamlit

MLOps & DevOps: Docker, Git, GitHub Actions, GitLab CI, FastAPI, MLflow, Evidently, Linux, VPS deployment

Spoken languages: French and Arabic (native), fluent English (preparatory class taught entirely in English)

Education

- Bachelor in Computer Science, AI & Data major** (3rd year, ongoing) · Ynov, Aix-en-Provence 2023 – present
Maths & data science (statistics, R, SAS, predictive modelling), SQL / NoSQL, BI, DevOps, Agile · hackathons, 48h Challenge, Ydays
- Engineering preparatory class** (taught in English) · South Mediterranean University, Tunisia 2022 – 2023
- Baccalaureate, Computer Science major**, with honours · Lycée 9 Avril, Sfax 2022